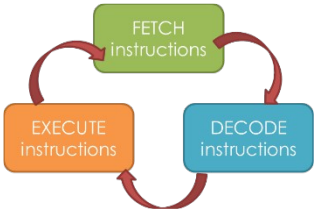
The Function of the CPU

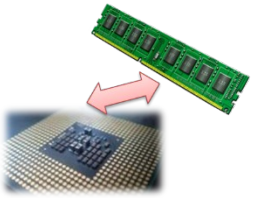
What is the Purpose of the CPU?

The CPU is often known as the 'brain of the computer'. Its job is to process data. And by processing we mean things like searching, sorting, calculating and decision making. Whenever you are on working on your computer, it is the CPU which is at the heart of everything.

The CPU follows three steps in order to process data. It is known as the Fetch - Decode - Execute cycle (aka Fetch-Execute Cycle).

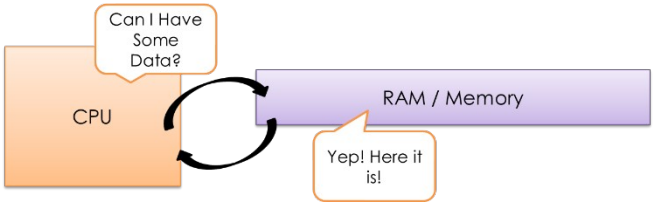


To begin with, whenever you open and work with a program, its data and instructions are loaded onto the RAM. As the RAM is accessed directly by the CPU, the CPU can get to work!



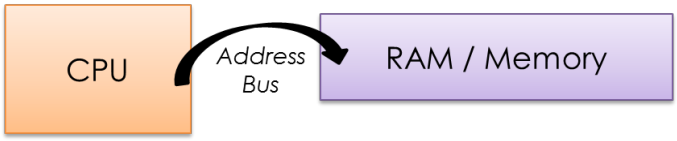
The Fetch Stage

In this step the CPU fetches some data and instructions from main memory (RAM) and then stores them in its own temporary memory called 'registers'.



For this to happen, the CPU uses a piece of hardware path called the 'address bus'.

The address of the next item that the CPU wants is put onto the ‘address bus’.



Data from this area then travels from the RAM to the CPU on another piece of hardware called the ‘Data Bus’.



The Decode Stage

Computer Science UK Membership Site Licence: Do not share outside your centre.

The decode step is where the CPU understands / works out what the instruction it has just fetched actually means. The CPU 'decodes' the instruction and gets things ready for the next step.



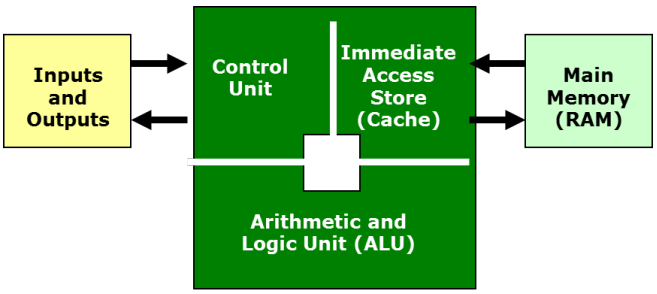
The Execute Stage

The Execute stage is where data processing happens. Instructions are carried out on the data. Once a cycle has been completed, another begins.

What makes up a CPU?

Now that we know the basic actions performed by the CPU it's time to look at the areas of the CPU responsible:

- Control Unit
- Immediate Access Store (IAS) or "Cache"
- Arithmetic and Logic Unit (ALU)



The Control Unit

There are three main jobs of the Control Unit:

- It manages and monitors hardware on the computer to ensure the correct data goes to the correct hardware.
- It manages the input and output signals ensuring these are dealt with correctly.
- It uses clock signals to synchronise the running of the Fetch-Decode-Execute cycle.

Immediate Access Store (Cache)

This part stores the data which is to be immediately processed. The CPU takes a chunk of data / instructions from the RAM and keeps it close so that it always has a constant supply of data to process. If data and instructions were moved from the RAM one instruction at a time, the CPU would work far slower because the CPU cycles much faster than the RAM can deliver data. So instead, chunks are moved from the RAM and stored on the cache so the CPU doesn't spend wasted time waiting for a delivery of data. Efficiently coded programs will also ensure that regularly used instructions are stored in the cache so that they can be accessed regularly, at speed.

Arithmetic and Logic Unit (ALU)

This is where the CPU actually carries out the maths and logic on the data (processes it).

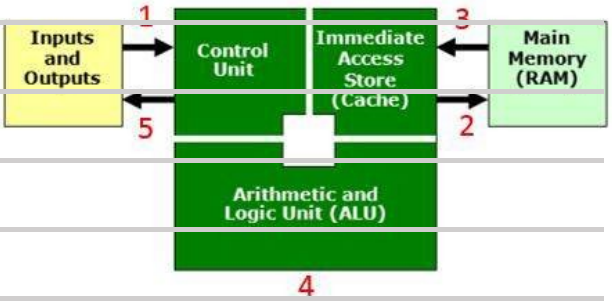
Computer Science UK Membership Site Licence: Do not share outside your centre.

It has two parts:

- Arithmetic part, which performs calculations on the data, e.g. 3 + 2 = 5
- Logic part, which deals with logical operations such as is True / False / Equal to / Greater than etc.

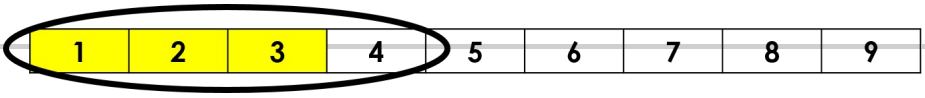
Summary of the journey of data / instructions around the CPU

1. An input device (e.g. keyboard) sends data to the CPU. The Control Unit receives this data.
2. The Control Unit sends this data into main memory to be used later.
3. When the time is right, the data will be transferred from main memory into cache (IAS)
4. The data will then be sent to the ALU for processing
5. The control unit will send the processed data back (for example to an output device such as a screen or monitor).



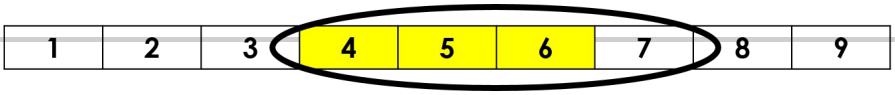
Questions *(The question zone you choose must either match your target grade or be higher!)*

Question Zone 1-3



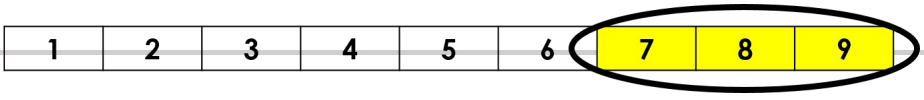
1. State the names of the 3 main components that make up the CPU. [3]
2. Define the words 'decode' and 'execute', in relation to the function of the CPU. [2]
3. Describe what happens during the F-D-E cycle. [3]

Question Zone 4-6



1. Describe what happens during the F-D-E cycle. [3]
2. Describe the roles of both the address and data bus. [2]
3. Explain the job of the CPU's control unit. [6]

Questions Zone 7-9



1. Explain the job of the CPU's control unit. [6]
2. Explain, using a diagram, the journey of an instruction from input device to output device. [5]
3. Discuss differences between the roles of both the RAM and the cache in how data is delivered to the CPU, during the fetch-decode-execute cycle. [4]

Computer Science UK Membership Site Licence: Do not share outside your centre.

Checklist:

- Date and title, clearly presented
- Spelling & grammar checked
- Question numbers in the margin
- Handwriting neat & legible
- Punctuation / Capital letters

Keywords / Key Terms:

Buses: *The communication channels linking the CPU with the RAM and I/O devices.*

Execute: *The action of a CPU performing an instruction.*

Decode*: Making sense of an instruction.*

CPU*: Central Processing Unit*

Computer Science UK Membership Site Licence: Do not share outside your centre.

Stick answer sheet here

State/Identify/Give/Name: Simply label a diagram, list, give a name or write a few words
Describe: Describing is 'saying what you see' (E.G.: A computer will have a CPU, Primary and Secondary storage etc)
Explain: Explaining is 'saying WHY/HOW something is like that'. (E.G.: A computer will have a CPU so that it can process all of the data the computer needs to perform a range of tasks. Primary and Secondary storage is needed because...)
Discuss: Discussing is 'looking at two sides of an issue, weighing up the two views and giving a conclusion'. Often these require a mini essay answer. (E.G.: New technology could be seen as being bad for the environment because..., but on the other hand, new technology has led to... In conclusion I believe that...)
Describe/Explain/Discuss **using examples**: Finally, if you are asked to give examples in any of these types of questions – YOU MUST **GIVE EXAMPLES**!

Computer Science UK Membership Site Licence: Do not share outside your centre.

Reflections **Score:** / **Percentage:** % **Grade:** **Progress** On / Above / Below

What Went

- I demonstrated a good level of understanding.
- I responded to the command words effectively.
- My answers were detailed / were written in depth.
- My work was well presented / legible.
- My answers effectively incorporated technical terminology.
- My responses were well structured / organised.
- My revision strategy was effective as I showed depth of understanding in my answers.
- My answers contained enough points / examples / explanations to achieve the marks available.

Even Better

- My answers need to be more accurate.
- I must respond correctly to the command words.
- My answers need more detail / greater depth.
- I must take greater care over my work / write neatly.
- I must incorporate key terminology into my answers.
- I must better organise my answers to improve its clarity.
- I need to improve my revision strategy as I did not demonstrate a depth of understanding in my answers.
- My answers didn't contain enough points / examples / explanations to achieve the marks available.

Further thoughts:

_____

_____

_____

_____

Computer Science UK Membership Site Licence: Do not share outside your centre.